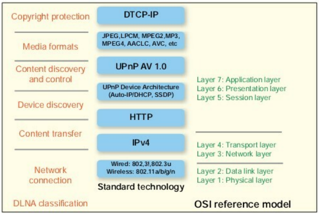

Figure 1: Protocol stack of DLNA (Image courtesy sony.net)

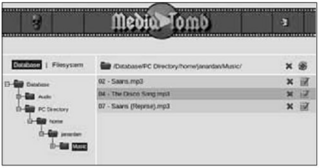

Figure 2: Media Tomb server settings

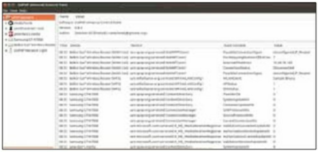

Figure 3: GUPnP Universal Control Point

'everything's automatic'. But rarely will you find, in one place, what equipment is required, and how to configure DLNA? Is it like Bluetooth, with which you pair devices before connecting, or like Wi-Fi, where you discover or manually enter an SSID to connect? Understanding the DLNA architecture helps you find the answers faster than the hours spent searching the Internet.

Since DLNA relies on UPnP, it is like the physical plug-and-play experience—remember Microsoft's Plug-and-Play feature from Windows 95 onwards? UPnP tries to offer the same experience in the digital media world. Figure 1 compares the OSI reference model to DLNA terms. The physical and data-link layer are the IEEE Ethernet and Wi-Fi standards, so your network can be wired or Wi-Fi (smartphones, tablets, etc) based. UPnP is a layer over the IP network, which discovers and communicates between your Wi-Fi or Ethernet-connected home devices. This is the important layer that really gives you the plug-and-play experience, and also selects the media types that can be supported on the network.

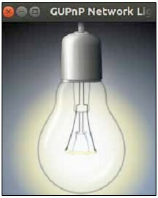

Figure 4: UPnP Router Control

Open source software—servers, clients and renderers

Consumer electronics equipment such as TVs or DVD players come with DLNA capability embedded, and will discover each other without manual configuration—but if you want to set up your PC as a DMS, then you need to do some configuration first. One DMS software that will make the videos, pictures and music on your PC available to all DLNA compatible players is Media Tomb— it is simple, easy to configure, and has been released under the GPL. After installing it, the server runs as part of system services. To enable a browser-based UI to add or remove media, follow the steps indicated below. First, as the superuser, edit */etc/mediatomb/config.xml* and if any of the options below are *'No'*, change them to *'Yes'*:

```
<ui enabled="yes" show-tooltips="yes">
  <accounts enabled="yes" session-timeout="30">
    <account user="mediatomb" password="mediatomb"/>
  </accounts>
</ui>
…
<transcoding enabled="yes">
```

After that, do a *sudo /etc/init.d/mediatomb restart* to activate the new configuration. Then, in your browser, go to *http://localhost:49152/* (assuming you haven't changed the default port). Log in with the name and password: *mediatomb/mediatomb* and you can then add the directories or files you would like to share on the home network to the database (see Figure 2).

Other DMSs include miniDLNA and Rygel. The former is used as the media server, in embedded form, by products like Seagate NAS storage, for home use. The XBMC 'software media player and entertainment hub' also comes as a live CD and can make your PC a dedicated DMS.

DLNA/UPnP-related software

The GUPnP project has many applications to help you discover, monitor and track UPnP devices. GUPnP Universal Control Point discovers all UPnP-capable devices on the network. In